

Chapter Six Prompt Writing for Specific Applications (Specialization)

6-1: Text Content Production (Article, Caption, Script, Sales Email)

Every type of text content has a different "brain" and "heart." A good prompt for content must specify the **final goal**. Let me build you a golden table:

Content Type	Main Goal	Key Verb in Prompt	Must Have
Blog/SEO Article	Inform and increase traffic	"Write a comprehensive and SEO-optimized article"	Keywords, H2/H3 headings, inviting conclusion
Instagram Caption	Engagement (likes and comments)	"Write a catchy and engaging caption"	Hook in the first line, closing question, relevant emojis
Script/Teaser	Convey emotion and message in a short time	"Write a script for a 30-second teaser"	Scene description, dialogue, voiceover
Cold Sales Email	Convert audience into customers	"Write a cold and persuasive email"	Catchy subject, audience's problem, call to action (CTA)

Golden example for an Instagram caption (since it's the most common):

*"As a professional copywriter, write an Instagram caption to introduce a **freshly roasted coffee brand**. The audience is young people aged 20 to 30. **[Format]** The caption should have 3 parts: 1) a hooking and surprising sentence about the smell of coffee, 2) 3 lines about the freshness of the beans, 3) a question to get comments. **[Tone]** Intimate, casual, yet luxurious. **[Constraint]** Maximum 150 words and with 5 relevant hashtags."*

6-2: Programming and Debugging (Troubleshooting, Code Optimization, Language Conversion)

Here AI is the best coding assistant in the world, provided you specify exactly **what language, what inputs, and what output** you want. The biggest mistake is saying "write me code"; instead you should say: "As a backend developer, write code that solves problem X in language Y and handles errors Z."

How to write a programming prompt (4-line formula):

1. **Language and version:** (e.g., Python 3.11 or JavaScript ES6)

2. **Input and output:** (input is a list, output should be a dictionary)
3. **Constraint and performance:** (must run in less than 0.1 seconds or use little memory)
4. **Debugging:** If you have an error, **copy-paste the full error (Error Traceback)** and say which part of the code you don't understand.

Concrete example for debugging:

*"I wrote this Python code to read a CSV file, but I get the error **'KeyError: 'price'**".*

[Paste your code here]

*Please, as a senior programmer, **step by step** analyze why this error occurs, then write the corrected code and also give a 2-line explanation of how to prevent this error from happening again."*

6-3: Data Analysis and Excel (Writing Formulas, Analyzing Tables, Discovering Patterns)

If you work with Excel or Google Sheets, AI can write your complex formulas or even discover hidden trends in your data. You just need to describe the **data structure** to it (you don't need to give the whole file, just say what the columns are named and what type of data is in them).

Two golden applications:

1. Writing a complex conditional formula:

"I have a table with columns: 'Purchase Date', 'Amount', 'City'. I want to write an Excel formula that if the amount is over 1 million and the city is Tehran, returns 'VIP Customer', otherwise returns 'Regular'. Please explain the formula and how to write it."

2. Pattern discovery and prediction (mental analysis):

"My sales data for the last 6 months shows that every month in the third week, sales drop by 20%. As a data analyst, write 3 possible reasons for this drop in online cosmetics stores and suggest what I should do to compensate for it next month."

6-4: Creating Images with AI (Midjourney / DALL-E)

Prompt writing for images is completely different from text! Here you are a **"film director."** Image generation models don't understand words; rather, they translate them into **visual attributes**.

Magic formula for image creation = [Subject] + [Environment/Context] + [Art Style] + [Lighting] + [Camera Angle] + [Quality]

Let's build a practical table of keywords for each part:

Part	Practical Examples
Subject	"A lone warrior woman", "A space cat"
Environment	"In the heart of a misty forest", "On the surface of Mars"
Art Style	"In the style of Van Gogh's oil painting", "Cinematic", "Anime", "Photorealistic"
Lighting	"Golden hour light", "Moonlight", "Hard studio light"
Camera Angle	"Top-down close-up", "Wide shot", "Macro"
Quality	"8K resolution", "Highly detailed", "Octane render"

Example of a professional prompt for Midjourney (translated and adapted):

*"A **smart black cat** wearing sunglasses, sitting on a **cartoon chair in space**. **Style:** A blend of cyberpunk and Studio Ghibli anime. **Lighting:** Vertical blue and pink neon lights. **Angle:** Close-up of the face with a wide lens. **Quality:** Hyper-realistic, 8K, with visible fur texture."*

(Golden tip: In Midjourney, use parameters like `--ar 16:9` for cinematic, `--v 6` for the new version, and `--s 750` for more creativity.)

Combined Practical Project for Chapter Six (The Last Workshop Before the Final)

Now that you know all 4, I'll give you a real mission. Suppose you have designed **a mobile app for growing houseplants**. Write a prompt that forces AI to do these 4 things **simultaneously** (you can number the tasks in the same prompt):

1. **[Text]** Write an Instagram caption to introduce this app (happy and friendly tone).
2. **[Programming]** Write a simple Python function that calculates the number of waterings per week based on the plant type (cactus or fern).
3. **[Data Analysis]** Give me an Excel formula to calculate the plant's daily growth compared to the previous day.
4. **[Image]** Write a prompt for Midjourney to design a minimal and beautiful logo for this app (flat and modern style).

